

Aayush Nitin Maheshwarkar

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years of experience delivering large-scale, revenue-generating enterprise platforms. Proven expertise in Java 17, Spring Boot, and cloud-native microservices, processing 2.5M+ daily requests and contributing to \$600M+ in platform revenue. Deep experience in distributed systems design, multithreaded Java development, agentic AI pipeline engineering, and production hardening. Demonstrated initiative through independent side projects applying Spring Boot, Redis, JWT, and LangChain. AWS Solutions Architect certification in progress.

TECHNICAL SKILLS

Languages: Java (8, 11, 17), Python, SQL, Shell Scripting, C

Backend & Microservices: Spring Boot, Spring Cloud, Spring Security, Hibernate/JPA, REST APIs, Feign, JMS Messaging, Service Discovery (Eureka), Cloud-Native Architecture, Event-driven Architecture, Multithreading & Concurrency

Cloud & DevOps: Docker, Jenkins CI/CD, Gradle, Maven, Linux, Git | *Familiar:* AWS (EC2, ECS, S3, RDS, CloudWatch), Kafka, Kubernetes, Terraform

Security: OAuth 2.0, SSO, Okta, JWT, Spring Security

Databases & Caching: PostgreSQL, Oracle, MySQL, MongoDB, Redis, Caffeine (Loading Cache, refreshAfterWrite)

Observability: Log4j, Resilience4j (Circuit Breakers, Retry), Splunk, Dynatrace, Grafana, AppDynamics

AI & LLMs: LangChain, Claude, Microsoft Copilot, LLM instruction engineering, Prompt evaluation, RAG, AI governance

Practices: Agile (Scrum, Kanban), JUnit, Test Driven Development (TDD), Code Reviews, Production Incident Triage

PROFESSIONAL EXPERIENCE

Federal Express Corporation

Orlando, FL

Full Stack Developer III / Senior Software Engineer

Feb 2023 – Present

- Owned quality standards and final approval authority for Microsoft Copilot LLM instruction sets across multiple enterprise platforms, establishing evaluation criteria for clarity, completeness, security, and accuracy before production release.
- Contributed to a LangChain and Claude-powered agentic pipeline built to handle a projected 10x surge in errored shipment volume from 2,000 to 20,000 daily errored shipments; implemented database-level retry controls and authored Python scripts for bulk data cleanup and revenue recovery.
- Engineered Java 17 / Spring Boot cloud-native rating microservices processing 2.5M+ peak daily requests, including the Quote Platform pricing engine (\$450M billed / \$1.8BN quoted across 69 markets), contributing to \$600M+ in total platform revenue.
- Implemented dangerous-goods surcharge precedence logic governing stacking across country-specific rules, enabling migration of 34,000+ enterprise customers and unlocking \$41M direct / \$1.2B in linked revenue.
- Contributed to ANCHOR per-pack surcharge processing, generating \$208M in annualized incremental revenue through improved operational capacity and global scan coverage.
- Reduced rate-quote API latency by ~600ms via proactive Okta JWT pre-caching using Caffeine with async background refresh, eliminating synchronous token-fetch overhead.
- Implemented Caffeine Loading Cache across rating services, reducing downstream service calls at 2.5M+ daily request scale; mentored junior engineers on caching strategy and cache-candidate identification.
- Improved service reliability with Resilience4j circuit breakers, retry, and fallback patterns around all downstream API calls; Log4j structured logging for diagnostics.
- Built and maintained Grafana dashboards for real-time REST API traffic monitoring, tracking request throughput, error rates, and latency percentiles across 2.5M+ daily requests.
- Used AppDynamics APM to identify CPU, JVM memory, and resource bottlenecks across microservice nodes; surfaced health rule violations into Grafana via the AppDynamics data source plugin.

Federal Express Corporation

Full Stack Developer II

Orlando, FL

Jul 2021 – Feb 2023

- Led development of the Flexible Surcharge Assessment Service (FSAS), generating \$150M+ annualized revenue with an expected \$500M+ over five years.
- Contributed to PIECE initiative enabling granular multi-piece shipment rating, generating \$51M in annualized revenue.
- Built and deployed a production monitoring tool providing real-time server management visibility to operations management.

Rensselaer Polytechnic Institute

Graduate Teaching Assistant

Troy, NY

Aug 2019 – Dec 2020

- Mentored 300+ students across Computer Organization (C, MIPS, x86 assembly) and Intro to AI; conducted code reviews and supported HyperSlate proof construction.
- Co-authored peer-reviewed research at the AAAI 2020 Fall Symposium on AI for Social Good, analyzing UN/UNICEF/WHO datasets across 34 countries using Python and R.

IBM India Pvt. Ltd. (IBM Software Development Labs)

Software Engineer Intern

Bangalore, India

Jan 2019 – Jul 2019

- Conducted performance and capacity planning for CICS Transaction Gateway v9.2 on IBM z13 mainframe, measuring CPU cost per transaction, Transactions Per Second, zIIP offload, and memory/MEMLIMIT requirements across RESTful JSON webservice workloads.
- Authored IBM's official CICS TG v9.2 Performance Summary report, benchmarking v9.2 improvements over v9.1 using IPIC connectivity, SSL-secured clients, and COMMAREA/Channel payloads.

PERSONAL PROJECTS

ShipRate API | Java 17, Spring Boot, Redis, JWT, PostgreSQL, Docker, Maven

2026 – Present

- Architected a Spring Boot REST API simulating enterprise shipping rating logic, calculating base rates, fuel surcharges, and remote-area fees via a layered Controller–Service–Repository architecture mirroring FedEx production patterns.
- Implemented JWT-based authentication with Spring Security and OAuth 2.0, securing all rating endpoints with stateless token validation.
- Integrated Redis caching on surcharge lookup routes, reducing repetitive downstream computation and demonstrating distributed cache patterns at the service layer.
- Containerized with Docker and PostgreSQL; CI/CD via Maven and GitHub Actions; automated JUnit test suite.

AI Rating-Failure Triage Agent | Java 17, Spring Boot, LangChain4j, Claude API, PostgreSQL

2026 – Present

- Extended ShipRate API with an agentic triage module powered by LangChain4j and the Claude API, autonomously classifying and routing shipping rating failures based on error type and historical patterns.
- Implemented structured prompt engineering and RAG fundamentals: domain-specific system prompts, tool-use patterns for database queries, and retrieval of historical failure context to improve classification accuracy.
- Designed for the same error-correction problem space as the FedEx production pipeline, demonstrating AI-native engineering alongside the Java/Spring Boot stack.

ACHIEVEMENTS & PUBLICATIONS

FedEx Hall of Fame Award (FY24): The highest engineering recognition at FedEx, awarded for contributions to the Quote Platform initiative – \$450M billed / \$1.8BN quoted across 69 global markets in FY23.

AAAI 2020 Fall Symposium on AI for Social Good (Co-Author): “Socioeconomic and Geographic Variations Impacting the Spread of Malaria” – K-Means and hierarchical clustering across a 34-country dataset; identified a statistically significant negative correlation between GDP and malaria death rates. [\[GitHub\]](#)

EDUCATION

University of the Cumberland

M.S., Project Management | CGPA: 3.971/4.0

Williamsburg, KY

Jan 2024 – Aug 2025

Rensselaer Polytechnic Institute

M.S., Information Technology | CGPA: 3.8/4.0

Troy, NY

Aug 2019 – Dec 2020

PES University

B.Tech., Electronics and Communication Engineering (Minor: Computer Science)

Bengaluru, India

Aug 2015 – Jul 2019

LANGUAGES

Fluent in English, Hindi, and Marathi.